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Turkey Vulture Aggregations at a Water Barrier Provide Evidence of Later Migration and Increasing Population Size over 25 Years

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ABSTRACT.—At Rocky Point near the southern tip of Vancouver Island, British Columbia, Canada, Turkey Vultures (*Cathartes aura*) form aggregations (commonly called kettles) during the fall migration before crossing the Juan de Fuca Strait. We used daily estimated total Turkey Vulture counts from the Rocky Point Bird Observatory to assess whether the timing of the fall migration and the number of vultures in the kettles changed between 1999 and 2023. On average, migration lasted 24 d (5–95% of migrants) with the peak period (25–75% of migrants) lasting 10 d. Fall migration was delayed by an average of about 1.6 d/decade. Maximum kettle size increased by 4.2%/yr, resulting in a 2.8-fold increase over 25 yr. Counts of local resident Turkey Vultures present before migration increased by 3.1%/yr. Populations in eastern North America have shown similar changes in the timing of migration. In contrast, European soaring species have not changed their timing or are migrating earlier. Our data also agree with the widely observed increase in Turkey Vulture populations in North America based on Breeding Bird Surveys and hawk watches. Our study demonstrates that counts of migrating raptors in kettles at migration barriers can provide evidence of changing migration phenology and population counts, and that such counts may be a useful complement to hawk watches, Breeding Bird Surveys, Christmas Bird Counts, and eBird reports.

KEY WORDS: *British Columbia*; *Cathartidae*; *migration*; *phenology*; *population*.

LAS AGREGACIONES DE *CATHARTES AURA* ANTE UNA BARRERA ACUÁTICA APORTAN EVIDENCIA DE UNA MIGRACIÓN MÁS TARDÍA Y UN AUMENTO DEL TAMAÑO POBLACIONAL A LO LARGO DE 25 AÑOS

RESUMEN.—En Rocky Point, cerca del extremo sur de la isla de Vancouver, Columbia Británica, Canadá, los individuos de *Cathartes aura* se agrupan durante la migración otoñal antes de cruzar el estrecho de Juan de Fuca. Utilizamos los conteos diarios estimados de *C. aura* del Observatorio de Aves de Rocky Point para evaluar si el momento de la migración otoñal y el número de buitres en las agregaciones cambiaron entre 1999 y 2023. En promedio, la migración duró 24 días (5–95% de los migrantes), con un período pico (25–75% de los migrantes) de 10 días. La migración otoñal se retrasó en promedio aproximadamente 1.6 días por década. El tamaño máximo de las agregaciones aumentó un 4.2% anual, lo que resultó en un incremento de 2.8 veces a lo largo de 25 años. Los conteos de buitres residentes presentes antes de la migración aumentaron un 3.1% anual. Las poblaciones del este de América del Norte han mostrado cambios similares en el momento de la migración. Por el contrario, las especies

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planeadoras europeas no han cambiado su fenología o están migrando más temprano. Nuestros datos también coinciden con el aumento ampliamente observado de las poblaciones de *C. aura* en América del Norte, según los Censos de Aves Reproductoras y los conteos de rapaces migratorias. Nuestro estudio demuestra que los conteos de agregaciones en barreras migratorias pueden aportar evidencia de cambios en la fenología migratoria y en los tamaños poblacionales, y que dichos conteos pueden ser un complemento útil de los conteos de rapaces migratorias, de los Censos de Aves Reproductoras, de los Conteos Navideños de Aves y de los reportes de eBird.

[Traducción del equipo editorial]

INTRODUCTION

The phenology of numerous biological processes is changing in response to climate change (Inouye 2022). Prominent among these changes is the timing of bird migrations. In general, prebreeding (spring) migration is occurring earlier whereas post-breeding (fall) migration is more variable with some species migrating earlier, some later, and others not changing (Van Buskirk et al. 2009, Bitterlin and Van Buskirk 2014, Allcock et al. 2022, Steiner et al. 2022, Romano et al. 2023), but see Søderdahl and Tøttrup (2023) for a recent study finding more consistent delayed fall migration in European passerines. In raptors, there is substantial diversity in the changes to the timing of fall migration, both in Europe (Filippi-Codaccioni et al. 2010, Panuccio et al. 2016) and in North America (Therrien et al. 2017). Some of the variation in the timing of fall raptor migration appears to be related to migration distance, with a trend for long-distance migrants to migrate earlier or not change and short-distance migrants to migrate later (Filippi-Codaccioni et al. 2010, Van Buskirk 2012, Therrien et al. 2017). In addition, there is evidence for a positive association between increasing trends in population size and the magnitude of change in fall migration phenology of raptors and other soaring birds (Panuccio et al. 2016), a pattern that has also been found for the timing of spring migration in multiple European bird species (Møller et al. 2008, Saino et al. 2011). Several studies have found that migration patterns have not changed uniformly across phases of migration with early phases of fall migration often showing stronger changes than later phases (Van Buskirk et al. 2009, Panuccio et al. 2016, Allcock et al. 2022). However, the opposite trend was also reported by Van Buskirk (2012) where six of nine short-distant fall migrants, including especially the Bald Eagle (*Haliaeetus leucocephalus*) and Golden Eagle (*Aquila chrysaetos*), showed a greater delay in the 90th percentile than in the 10th percentile of the migration.

The Turkey Vulture (*Cathartes aura*) is a widespread species in the western hemisphere. The species is a partial migrant with populations that breed at high latitudes migrating long distances and those breeding closer to the equator tending to migrate less or not at all (Dodge et al. 2014, Kirk et al. 2024). As an obligate soaring species, Turkey Vultures are highly sensitive to changes in thermal updrafts, which are affected by temperature (Dodge et al. 2014, Mallon et al. 2021). Therefore, the timing of their migration may be particularly affected by climate change. Three studies of fall migration have found delays of about 1 d/decade in eastern and central North American populations (*C. a. septentrionalis*; Van Buskirk 2012, Therrien et al. 2017, Steiner et al. 2022), but only Steiner et al. (2022) included data from the last decade. To the best of our knowledge, there have been no published reports of changes in migration timing for western populations (*C. a. meridionalis*).

A broad range of studies agree that Turkey Vulture populations are generally increasing or stable in both eastern and western North America, with increases related to northward range expansion while some local or temporary declines in the west may be due to droughts (Kiff 2000, Smith et al. 2008, Rosenberg et al. 2019, Oleyar et al. 2021, Smith et al. 2023, Kirk et al. 2024). Some of the analyses are based on Breeding Bird Surveys that are not well suited to counting vultures and other soaring species because they take place early in the day before most individuals begin soaring (Mallon et al. 2020). Nevertheless, they may reveal trends because this bias is probably constant over time (Kiff 2000). Breeding Bird Surveys indicate increasing populations for the Pacific coastal region of Canada on both long-term (1970–2021) and short-term (2011–2021) time scales (Smith et al. 2023). Hawk watch counts, included in some analyses, reveal similar patterns (Smith et al. 2008) but also require cautious interpretation because of the potential for changes in the proportion of the population migrating or in the migration routes selected (McClure et al. 2022). Many of the published analyses are based on data only up to 2017 or earlier, leading to calls for more recent demographic data to allow

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conservationists to recognize potential problems early (Santangeli et al. 2022).

Migrating raptors, like other bird species, often avoid barriers such as mountain ranges and open water, leading to high concentrations as they funnel into areas where the barriers are absent and crossing conditions are more favorable (Bildstein 2006, Bayly et al. 2018). Open water is a particularly obstructive barrier for soaring species, including Turkey Vultures, because of the limited occurrence of thermals (rising warm air) in this environment (Santos et al. 2020, Nourani et al. 2021, Pekarsky et al. 2024). Turkey Vultures, like other raptors, may stopover at locations such as the tips of peninsulas that minimize crossing distances as they wait for favorable conditions to make the crossing (Bildstein 2006). At these bottlenecks, large numbers of birds often soar together in thermals to gain elevation; these formations are commonly called kettles (Bildstein 2006).

Hawk watches are often located at bottlenecks because counts at these locations can assess populations converging from a broad area. However, protocols for hawk watches exclude kettles to reduce the risk of multiple counts of the same individuals (Bildstein et al. 2007, Dunn et al. 2008). Nevertheless, kettles provide a conspicuous indication of the presence of migrants and thus can provide information about the timing of migration. In addition, kettles may also provide a relative measure of changes in population size. On average, changes in the size of the largest kettles would be expected to correlate with the number of migrants and therefore the size of the population, even though weather-related changes in the numbers at a bottleneck will also affect kettle size. Thus, kettles can provide information about changes in population size as well as migration phenology. The goals of our study were to use Turkey Vulture counts, primarily from kettles, recorded at Rocky Point on the southern tip of Vancouver Island, Canada, over the fall migration period from 1999 to 2023, to examine whether the timing of different phases of migration and the largest number of birds observed each year (and thus the underlying population size) had changed over that period.

METHODS

Study Site and Data Collection. Rocky Point Bird Observatory (RPBO) operates a fall bird migration monitoring station on the Canadian Department of National Defence's Rocky Point Ammunition Depot at the southern tip of Vancouver Island, BC, Canada, on the traditional lands of the Sc'ianew people (48.3187°N, 123.54672°W; Fig. 1). From the peninsula on which RPBO is located, the distance across the Juan de Fuca Strait to the Olympic Peninsula in Washington, USA, is approximately 18 km. Vancouver Island is the largest island on the west coast of the Americas, 450 km long and 100 km at the widest point, with the long axis oriented from northwest to southeast. The eBird status and trends map shows the breeding season distribution of Turkey Vultures ranging from the south to the north end of Vancouver Island and adjacent coastal mainland (Fink et al. 2023). There is greater abundance in the southern and eastern parts of the island as compared to the western, central, and northern parts of the island and the nearby mainland. Turkey Vultures form kettles over Rocky Point before crossing the strait to Washington. They also form kettles in East Sooke Park approximately 8 km to the west. Kettles that are usually smaller also occur near Victoria, approximately 22 km northeast. Recent tracking studies have indicated that individuals move among these locations and may leave Vancouver Island from any of them, including an alternate route via the San Juan Islands between Vancouver Island and the USA mainland (Gillian Radcliffe and Cara Herrington, The Raptors, Duncan BC Canada and University of Victoria pers. comm.). The relative use of these different locations and the conditions that may favor switches among them are not known.

We derived Turkey Vulture counts from data gathered at Rocky Point by RPBO as part of its avian monitoring program from 1999 through 2023. The field season lasted from 23 July or earlier to 18 October during which data were collected by a standardized protocol involving an early morning census, mist netting to capture passerine birds for banding starting 30 min before sunrise and ending 6 hr later, and general observations. Personnel present each day included a licensed bander and at least one other individual with excellent identification skills as well as an assistant and up to three additional assistants and trainees. Nets were checked at 20-min intervals, and the protocol required personnel to record as general observations all birds seen as they moved to or between nets. Although Turkey Vultures were seen during the early morning census, large kettles were observed primarily later in the day when thermals became available for soaring. The route to four nets, located along a 100-m stretch at the edge of the southeast meadow (48.3188°N, 123.5464°W), provided personnel with an unobstructed view of the sky and the Juan de Fuca Strait so that no kettles of Turkey Vultures were likely to be missed. After the nets were closed, all personnel gathered to determine

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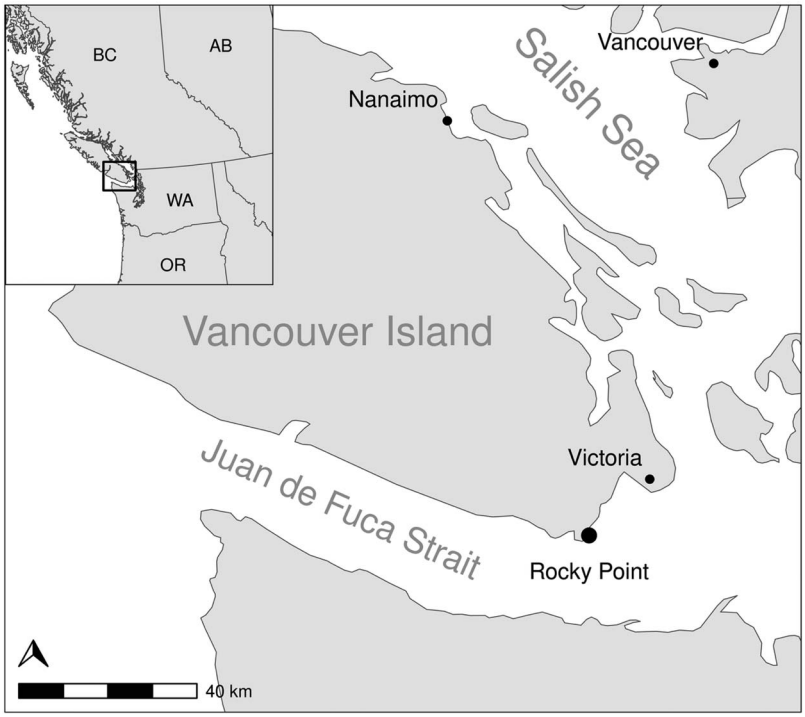

Figure 1. Map showing location of Rocky Point (large dot) in relation to Vancouver Island and the cities of Vancouver, Nanaimo, and Victoria (small dots), in British Columbia, Canada.

the daily estimated total (DET) for each species. The DET was the sum of birds captured in mist nets, counted in the census, and tallied during general observations, with every effort made to avoid double counting. For Turkey Vultures during the migration season, the DET was very similar to the maximum number seen during the latter part of the mist-netting period.

We did not include data from days when rain prevented use of the mist nets or when the nets were closed more than one hour early due to adverse weather. Because the station is located on a military base with restricted access, some days were also missed when access was denied. The first year of station operation (1998) using the described protocols had extensive closures during the migration period, so we did not include data for that year. In addition, we were denied access during the entire year of 2007. For the other 24 yr, during the migration period (28 August–18 October), 19% of the days lacked counts (70% due to military closure, 30% due to weather). Unfortunately, these blanks resulted in substantial gaps during the migration period in 2 yr, which we took into consideration in our analysis. Overall, our analysis is based on 24 observation years over a 25-yr period. To ensure a consistent record across years, we included surveys only on or after 23 July.

Statistical Analysis. All data summaries and analyses were performed in R (R Core Team 2025). To calculate metrics of migration timing and duration, we generally followed methods from Allcock et al. (2022) by first estimating migration phenology as a smoothed curve from which we could then calculate start, end, and peak of migration, while smoothing over effects of weather or unsampled days. We used general additive models (GAMs) to create these smoothed curves as they are flexible and appropriate models for when phenology patterns are not well known (Lindén et al. 2017). We fit GAMs to daily counts over the banding season for each year using the *mgcv* R package v1.9-0 (Wood 2017). The GAMs modeled DET by day of year as a negative binomial distribution, smoothing year with the default thin plate regression spline smoothing basis, and with restricted maximum likelihood. From these models, we extracted the predicted DET for each date, and these were then used

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to calculate metrics of migration timing and duration. The maximum number of local resident Turkey Vultures observed before the migration started was generally small (1–8) but occasionally could be larger due to an attractive food source such as the carcass of a marine mammal on the nearby shore. Therefore, we removed local resident individuals from the predicted fall migrant counts prior to calculating migration timing and duration in each year. To remove these individuals, we calculated the median predicted DET before migration onset (<28 August) for each year. We then subtracted this number from each subsequent predicted DET value in that year. We chose 28 August as a cutoff date as it was close to, but never overlapped, the start of migration in any year (Supplemental Material Fig. S1). We used these adjusted DETs (starting on 28 August) to create a cumulative distribution from which we calculated the dates of each phase of the migration using percent passage: the start of the migration (day on which 5% of the cumulative count occurred), start of the peak migration (25%), peak migration (50%), end of the peak migration (75%), and end of the migration (95%). We also calculated the duration of migration (day of 95% minus day of 5%), and the duration of peak migration (day of 75% minus day of 25%). Because these dates are calculated from fitted GAMs, they should not be unduly influenced by occasional days of missing data except when observations were missed for multiple days during the peak or at the end if substantial migration continued after the data collection period.

To assess whether migration dates and durations changed over time, we ran linear regressions of each phase of the migration (identified by percent passage as previously outlined) against year. Further, we assessed whether migration dates were changing at the same rate for the different phases. To this end, we ran a linear regression to look at how the interaction between year and phase varied with date.

To assess whether the number of migrating Turkey Vultures had changed over time, we used generalized linear models to model annual maximum raw observed DETs (local residents included) as a function of year. We modeled the maximum annual raw DET by year with a negative binomial distribution to account for overdispersion (using the *MASS* R package; v7.3-60; Venables and Ripley 2002). To assess whether the number of local resident Turkey Vultures present before migration started had changed over years, we modeled the median observed number of local resident Turkey Vultures (before 28 August) by year with a Poisson distribution. To facilitate interpretation of the results, we exponentiated the model estimates, permitting us to interpret the estimates as percentages.

Model assumptions were assessed by reviewing simulated residual diagnostic plots with the *DHARMa* R package v0.4.6 (Hartig 2022), and sensitivity was assessed by reviewing Cook's distances (Cook and Weisberg 1982). Years with high Cook's distance values (>4/sample size) were systematically excluded from the analyses, one at a time, to determine their potential influence on the analysis results. No removal of any year with a high Cook's distance changed the interpretations of the analyses. We used an alpha of 5% for statistical significance. All code and data to reproduce these analyses are available on GitHub and through Zenodo at https://github.com/steffilazerte/vulture_migration (https://doi.org/10.5281/zenodo.13830989).

RESULTS

The median date of the estimated peak migration was 28 September (range = 25 September–4 October). The median start date of migration was 17 September (10–22 September) and the median end of migration was 11 October (4–16 October). The entire migration (5–95% passage) lasted an average of 24 d (range = 16–30 d), while the peak migration (25–75% passage) lasted an average of 10 d (range = 6–11 d). The GAMs for each year with daily counts as well as the peak, estimated start and end of the migration, and start and end of the peak period are presented in Figure S1.

Peak migration was significantly delayed by an average of 1.6 d/decade between 1999 and 2023 (Table 1, Fig. 2A). The dates of the start of migration as well as the start and end of peak migration were also delayed as indicated by positive, statistically significant slopes. The slope for the end of migration was also positive but not statistically significant. The estimated slopes gradually decreased from the start of migration (2.1 d/decade) to the end (1.2 d/decade), but these differences were not statistically significant (interaction between year and phase in the secondary model; $F_{132,\,5} = 0.211$; $P = 0.957$). There were no significant changes in the duration of migration or peak migration (Table 1). To check for possible effects of data gaps, we repeated the analyses after removing 2 yr in which there were sampling gaps around peak migration (2011, 2013) and found that there was no substantial effect on slopes or significance levels. Nevertheless, it is worth noting that the late peak in 2011 (Fig. 2A) occurred 4 d after a 10-d military closure.

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Table 1. Slopes (plus SE, *T*-statistic, and *P*-value) of linear regressions relating day of the year on which each of five phases (defined by percent passage) of the fall migration occurred to observation year ($n = 24$ yr) over a 25-yr period from 1999 to 2023 for Turkey Vultures at Rocky Point, BC, Canada. The slopes, SE, *T*-statistics, and *P*-values are shown for the relationships between duration of migration and duration of peak migration and observation year.

Migration Phases and Durations (with % Passage)	Slope (d/Decade)	SE	*T*-statistic	*P*-value
Start (5%)	2.06	0.74	2.784	0.011
Start of peak (25%)	1.78	0.54	3.305	0.003
Peak (50%)	1.64	0.56	2.920	0.008
End of peak (75%)	1.56	0.58	2.697	0.013
End (95%)	1.19	0.84	1.146	0.171
Duration of migration (95%–5%)	−0.87	0.96	−0.909	0.373
Duration of peak migration (75%–25%)	−0.22	0.37	−0.588	0.563

Removing 3 yr in which migration may have continued after the end of the field season (2001, 2006, 2023) resulted in similar slopes and statistical significance. Even removing all 5 yr resulted in only minor changes in the significance of some slopes despite the reduced sample size (Table S1).

The maximum annual DET of Turkey Vultures, including local residents, averaged 831 individuals

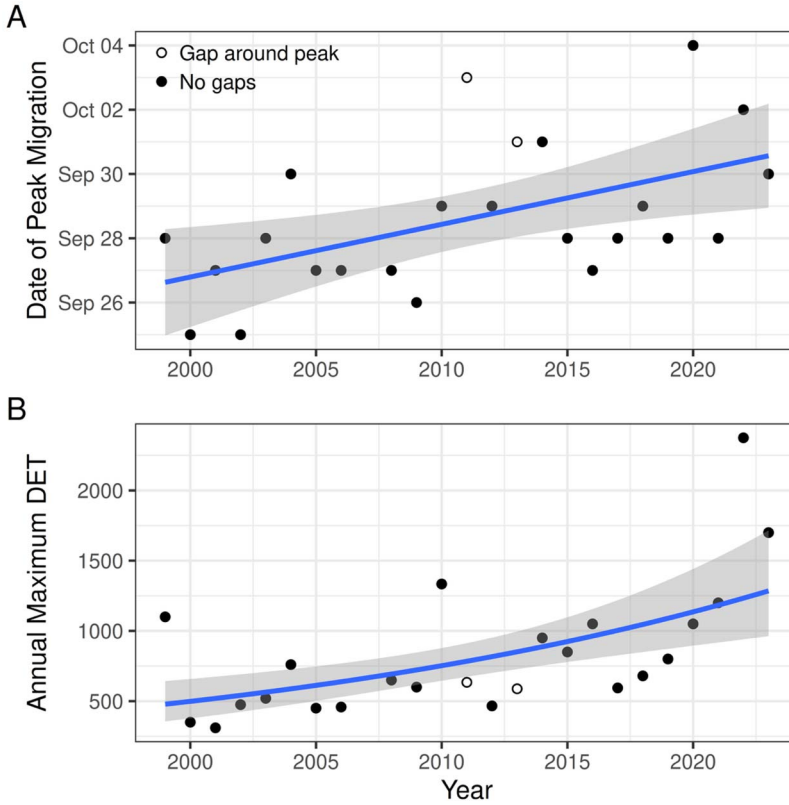

Figure 2. Relationships between (A) the date of peak fall migration and year of observation and (B) the annual maximum raw daily estimated total (DET) and year of observation for Turkey Vultures at Rocky Point, BC, Canada, between 1999 and 2023. Points and open circles represent individual years ($n = 24$), with the open circles indicating 2 yr in which there were sampling gaps around the time of peak migration. (A) Line shows the linear regression. (B) Line shows the negative binomial generalized linear regression. Both panels show the 95% confidence interval as a gray band.

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(range = 310–2375). The lowest DET was in 2001 and the highest DET in 2022 (Fig. 2B). The number of Turkey Vultures significantly increased over the observation period (4.2%/yr, $Z = 3.88$, $P < 0.001$). This annual increase is equivalent to about a 2.8-fold increase over the 25-yr period. Repeating the analysis of annual maximum DET without the 2 yr with sampling gaps near the peak, without the 3 yr in which migration appeared to be continuing after the banding station had closed, or with all 5 yr excluded resulted in only very minor effects on the slope (Table S1). In addition to exploring the effect of gaps in the observations, we also checked the potential effect of extreme values as indicated by Cook's distances. Visual inspection indicated that 2022 was a very high year with the potential to drive the patterns in this model and also had a relatively high Cook's distance ($D = 0.53$). However, when removing it (as part of our sensitivity checks on all points with high Cook's distances in all analyses), the increase in numbers remained highly significant ($P = 0.001$; $Z = 3.182$), although the slope was lower (3.4%/yr). The median number of local resident Turkey Vultures observed prior to the start of migration averaged 5 individuals (range = 1–8). This number also increased over the observation period by an average of 3.1%/yr ($Z = 2.409$, $P < 0.02$).

DISCUSSION

Despite substantial variation in the timing of migration among years, there was a clear trend for migration to occur later over the 25-yr observation period, whether measured at the start of migration, peak migration, or the start or end of the peak migration period, although the end of migration did not change significantly. To the best of our knowledge, this is the first evidence for a change in the timing of Turkey Vulture fall migration for western North America.

There is some evidence that passerines migrating long and short distances have changed the timing of migration in different ways (Jenni and Kéry 2003, Søderdahl and Tøttrup 2023). Long-distance migrants are more likely to have started their fall migrations earlier or not changed much over time whereas short-distance migrants were more likely to have delayed their migration or delayed it to a greater extent. This pattern also applies to raptors in both Europe and North America (Filippi-Codaccioni et al. 2010, Panuccio et al. 2016, Therrien et al. 2017, Steiner et al. 2022). Turkey Vultures from northern North America are long-distance migrants (Dodge et al. 2014, Kirk et al. 2024). However, they appear to be exceptions to the general pattern because our study, as well as research from eastern and central North America, has found fall migration occurring later rather than earlier (Therrien et al. 2017, Steiner et al. 2022). Delayed migration may be related to changes in climate that allow conditions to remain suitable for foraging and/or for migration later in the fall. Alternatively or in addition, climate change may create conditions that facilitate spending the nonbreeding period closer to the breeding locations, thus allowing a shorter migration that could start later (Visser et al. 2009).

Studies on the phenology of other species have found inconsistent directions of effect for the start, middle, and end of the migration; even when effects were in the same direction, the magnitude differed greatly (Van Buskirk 2012, Panuccio et al. 2016, Allcock et al. 2022). In our study, however, all phases showed a delay in migration, although the trend was not statistically significant for the end of migration. The trend for earlier phases to show a stronger delay than later phases suggests that eventually the migration duration could be reduced, but our data did not indicate that the duration of either migration or peak migration had decreased. A study of fall migration phenology of raptors in north-central North America found that Turkey Vultures did not show a difference among the phases of migration (Van Buskirk 2012). However, in contrast to our findings, the end of fall migration was more delayed than the start for many other species in that study. A study of European raptors and storks that mostly started their migrations earlier found that the early phases of migration tended to change more than the later phases (Panuccio et al. 2016). But in Garden Warblers (*Sylvia borin*) migrating through Poland, the shift toward earlier fall migration was stronger for the later stages of migration than for the earlier stages (Pinszke and Remisiewicz 2023). There appears to be enough potentially interesting variation in the phenology of different phases of migration to warrant consideration in future studies. Differences among phases may be related to different response of adults and juveniles to a changing climate. For example, Panuccio et al. (2016) suggested in their study of raptors and storks that adults often migrate earlier than juveniles and are able to adjust their timing based on experience while later-migrating juveniles without migration experience are less able to do so. Alternatively, if the migratory passage comprises multiple sub-populations that migrate at different times, changes in the timing of

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different phases of migration could be a result of complex interactions (Van Buskirk et al. 2009).

We found a substantial increase (about 4% per yr) in the annual maximum number of Turkey Vultures recorded over the observation period. This increase most likely reflects an increase in the size of the migrating population, because population size limits the maximum size of kettles. However, kettle size varied substantially around this trend from 310 to 2375 (about 7.7-fold) among years compared to the 2.8-fold increase modeled over this same period. It is important to note that our measure does not represent the total population because migration is spread over 16–30 d, making it likely that many birds migrated before and after this peak number. Some of the variation in kettle size may have been due to variation in the weather around peak migration. For example, if wind and thermals allow soaring over land but are not suitable for soaring over water, vultures may accumulate at the southern tip of Vancouver Island, increasing kettle size. However, if conditions favor crossing when birds arrive at the southern tip of the island, kettle sizes may remain smaller. Similar weather effects have been documented elsewhere for Turkey Vultures (Mallon et al. 2021) and other species (Santos et al. 2020). Nevertheless, our highest count (in 2022) occurred in an exceptionally hot, dry year that would appear to have been very suitable for Turkey Vultures and was not particularly late. In addition to weather-related variation in crossing at Rocky Point, changes in the location of kettles or the proportion of the population using different locations, induced by the weather or other factors such as food supply or habitat changes could have affected the numbers of birds observed in kettles at Rocky Point. Such variation is unlikely to alter the conclusion that there has been a substantial rise in numbers. However, if crossing conditions have changed systematically over time, the estimate of the rate of increase could be affected. Finally, there have been observations of some hundreds of birds crossing from the mainland to Vancouver Island via smaller islands north of Nanaimo (Fig. 1), suggesting that the Turkey Vultures counted at Rocky Point may include some from the mainland coastal area northwest of Vancouver (G. Monty pers. comm.). Thus, increases in mainland coastal populations, for example by northward range extension, or interannual variation in movement to the island could influence the trend seen on southern Vancouver Island.

Regardless of the potential confounding issues outlined above, our findings are in keeping with broad reviews that consider Turkey Vulture populations to be generally stable or increasing (Kiff 2000, Rosenberg et al. 2019, Oleyar et al. 2021, McClure et al. 2022, Kirk et al. 2024). Using Breeding Bird Survey data, Kirk and Hyslop (1998) concluded that Turkey Vulture populations in Canada increased by about 11%/yr between 1985 and 1994, but there were no data from western Canada included in the analysis. More recently, the Canadian Breeding Bird Survey results for 2011–2021 indicated an increase of 4.6%/yr for Canada, 6.6%/yr for British Columbia, and 5.9%/yr for the Pacific coastal region (Smith et al. 2025). Counts from hawk watches at six sites in the western United States up to 2001 also showed increases at the four that had the longest data series, although none of the sites were on the Pacific Coast Flyway (Hoffman and Smith 2003). For the western United States, data up to 2005 indicated increases of about 4–18% for two sites on the Intermountain Flyway and about 5% for two sites on the Pacific Coast Flyway, although annual numbers at the latter sites were quite low compared to ours (31–302 individuals per yr; Smith et al. 2008). The eBird data indicated that Turkey Vulture observations in British Columbia increased 21% from 2012 to 2022 (Fink et al. 2023); this implies an average annual increase of about 1.8%. For Vancouver Island, there is also an increasing trend. The finest scale provided by the eBird maps consists of 27 × 27-km areas. For the 54 such areas on Vancouver Island, the trend is increasing for 49 (three of which are statistically significant) and decreasing for five (none are statistically significant; Fink et al. 2023). Thus, our study of counts at a bottleneck and barrier supports evidence of increasing populations from Breeding Bird Surveys and eBird checklists; and shows a similar rate of increase to that found by Breeding Bird Surveys.

To the best of our knowledge, our study is the first to use kettle sizes to examine historical trends in migration timing or population size in a raptor. Our approach has some advantages over alternative methods. Unlike the Christmas Bird Count, it allows count estimates for fully migratory species as they leave breeding areas. Unlike Breeding Bird Surveys, the timing of daily counts can be adjusted to the time of day that most individuals will be visible. Using a standardized protocol makes the data more consistent than eBird reports. Counts of kettles can be carried out in less time than hawk watch counts at similar locations. However, limitations must also be recognized. Unlike hawk watches, counts of kettles do not provide an estimate of the absolute number of migrating birds. Even when kettles are counted at a standardized time of day, numbers may be reduced by favorable crossing conditions

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occurring before the count period or increased by preceding weather that allows movement over land but not across a water barrier. Furthermore, like hawk watches, kettle counts may be influenced by weather-related changes in migratory pathways. Given the clear trends observed in our study, it would be useful to compare kettle counts to hawk watch counts at the same location. Our study has shown that counts of kettle sizes can indicate inter-annual changes in both phenology and abundance in a soaring species and may thus offer another tool to better understand the complexity of migration and how it is responding to climate change.

SUPPLEMENTAL MATERIAL (available online). Figure S1. Daily estimated total (DET) number of Turkey Vultures at Rocky Point in the fall of each year from 1999 to 2023 (except 2007). Table S1. Effect of removing selected years on the estimates (and *P*-values) of slopes of the relationship between date of each phase of the Turkey Vulture migration at Rocky Point and year, of the duration of migration and peak migration and year, and of the annual maximum raw daily estimated total (DET) number of Turkey Vultures and year.

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